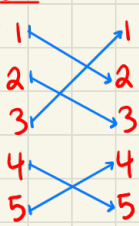
feb. 24

- $S_n = \{$bijective functions from a set with n elements to itself; group under function comp.$\}$

Ex.

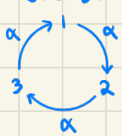
α



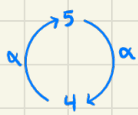
Cycle notation of α (write α as a product of disjoint cycles):

Composition

$(312) = (123) = (231)$



$(45) = (54)$



$(123)(45) = (45)(123)$

★ usually leave off elements that map to themselves

What is $|\alpha| = ?$

$\alpha^2 = \alpha \circ \alpha = (123)(45)(123)(45) \overset{?}{=} (123)(123)(45)(45) = (123)^2$ $(4)(5)$

$\alpha^3 = \alpha^2 \circ \alpha = (123)^2(123)(45) = (123)^3(45)^3 = (45)(1)(2)(3)$

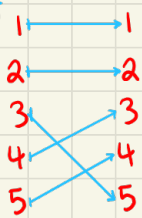
continue for α^4, α^5, etc, but that's a lot of work

$|(123)| = 3$, $|(45)| = 2$ 3-cycle has order 3, 2-cycle has order 2

$LCM(2,3) = 6$, so $|\alpha| = 6$

(earliest time both cycles return to id)

γ



$\gamma = (1)(2)(354) = (354)$

- Conjecture: Disjoint cycles commute?

- Thms. of Ch. 5
  - Thm. 5.1 Every permutation in S_n can be written as a cycle or product of disjoint cycles
  - Thm. 5.2 Disjoint cycles commute i.e if $\alpha = (x_1 x_2 \dots x_k)$ & $\beta = (y_1 y_2 \dots y_m)$ have no entries in common, then $\alpha \circ \beta = \beta \circ \alpha$ (also written $\alpha\beta = \beta\alpha$)

Warning: for $n > 2$, S_n is not Abelian

Counterexample: $\overset{\alpha}{(12)}\overset{\beta}{(13)}$ & $\overset{\beta}{(13)}\overset{\alpha}{(12)}$ $\quad \alpha\beta(\overset{\text{input}}{1}) = 3$

$\downarrow \qquad \downarrow$

$(132) \neq (123)$ $\quad \beta\alpha(\overset{\text{input}}{1}) = 2$

  - Thm. 5.3 The order of a permutation is the least common multiple of the lengths of its disjoint cycles
  - Thm. 5.4 Every permutation in S_n for $n > 1$ is a product of 2-cycles ← transpositions

Ex. (12479)

$= (19)(17)(14)(12)$ start w/left, correct for each position

id as product of 2-cycles: $(12)(12)$ since $n > 1$, so at least 2 elements

  - Thm. 5.5 The parity of a 2-cycle decomposition is unique (even or odd)

★ note: 2-cycle decompositions are not unique

- $A_n = \{\alpha \in S_n : \alpha$ has an even 2-cycle decomposition$\}$ is a subgroup of S_n, called the alternating group

Permutation Groups

Abstract Algebra

For this worksheet we will use cycle notation. For example, the permutation in S_4 such that $1 \mapsto 3$, $2 \mapsto 4$, $3 \mapsto 2$, $4 \mapsto 1$ is written (1324).

1) Write $\overset{\beta}{(15432)}\overset{\gamma}{(37126)}\overset{\sigma}{(23)}\overset{\varepsilon}{(45)}$ as a product of disjoint cycles.

$\alpha = \beta \circ \gamma \circ \sigma \circ \varepsilon$ follow elements this way

$\alpha(1) = \beta(\gamma(\sigma(\varepsilon(1)))) = \beta(\gamma(\sigma(1))) = \beta(\gamma(1)) = \beta(2) = 1$

etc ⋮

$\alpha = (1)(27536)(4) = (27536)$

2) Write $\alpha =$ (37126)(45)(15432)(23) as a product of disjoint cycles.

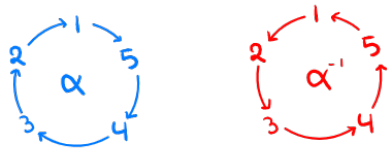
$2 \to 3 \to 2 \to 6$, $6 \to 3$, $3 \to 2 \to 1 \to 2$ $\Big\}$ (263)

$5 \to 4 \to 5$ $\Big\}$ (5)

$4 \to 3 \to 7$, $7 \to 1$, $1 \to 5 \to 4$ $\Big\}$ (471)

$\alpha = (263)(471)(5)$
$= (263)(471)$

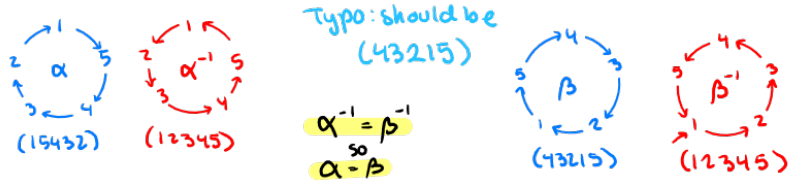
3) Compute the inverse of (15432).



$\alpha^{-1} = (12345)$

4) Show that $\overset{\alpha}{(15432)} = \overset{\beta}{(43251)}$ by showing they have the same inverse.

Typo: should be (43215)



$\alpha^{-1} = \beta^{-1}$
so
$\alpha = \beta$

5) Theorem 5.3 states that the order of a permutation written as a product of disjoint cycles is the least common multiple of the lengths of the cycles. Verify this by computing the order of $\alpha = (12)(354)$, $\beta = (12)(123)$, and $\gamma = (123)(456)$. Remember to write permutations as a product of disjoint cycles.

Worksheet has a back side, didn't get to it in class